

Lab Experiments 4 and 5

You are asked to implement Feed Forward Network and to use gradient descent with backpropagation for a classification task. You are not allowed to use any automatic differentiation packages. Perform all the operations using numpy for all matrix/vector operations. This network will be trained and tested using the MNIST dataset. The MNIST database of handwritten digits divided into 10 classes from 0 to 9. The dataset is included in the accompanying zip file ("data.txt" and "label.txt"). Randomly, divide the data into train, validation, and test splits using a 60, 20, 20 ratio.

You will experiment with different number of hidden layers with different number of neurons in each layer.

- (i) Using one hidden layer with 300 number of neurons.
- (ii) using two hidden layers with each layer is having 200 neurons.
- (iii) Using three hidden layers with each layer is having 200 neurons.

Run the experiments for 100 epochs, take learning rate $=0.01$. Use sigmoid as an activation function for the hidden layers and softmax as an activation function for the output layer. Later, can also experiment with different activation function (tanh) for the hidden layers.

Include plots of the training and validation error as a function of epochs in all the cases. Also mentioned the test accuracy at the end in all the cases.